

Department of Software Engineering

Constraints Applied on the Rent-a-Car Management System

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Rent-a-Car Management System

Introduction

A database constraint is a rule applied to a table or column to ensure the accuracy, consistency, and integrity of data stored in the database. In the Rent-a-Car System, constraints help prevent invalid records, maintain relationships between tables (Models, DAOs), and enforce operational business rules.

This assignment describes the constraints necessary to implement in the Rent-a-Car database layout and explains where each constraint has been, or should be applied per the existing system queries.

1. PRIMARY KEY Constraint

Purpose

A PRIMARY KEY uniquely identifies each record in a table. It does not allow duplicate values or NULL values.

Applied Tables

Customers Table

Reason: The system uses CNIC as the unique identifier for renting cars.

Vehicles Table

Reason: Ensures every car in the fleet is uniquely identified.

Drivers Table

Reason: A unique identity assigned to each driver.

Sales Table

Reason: Uniquely tracks every individual rental transaction logic.

Payment Table

Reason: Every money transfer must be uniquely tracked.

Expenses Table

Reason: Essential for cataloging all business outgoings.

2. FOREIGN KEY Constraint

Purpose

A FOREIGN KEY maintains referential integrity between related tables.

Applied Relationships

Sales Table

Reason: A sale (rental agreement) must belong to an existing registered customer, and checking out a valid vehicle from the fleet.

Sales Table (Optional Driver)

Reason: If a customer does not choose "Self-drive", the driver requested must exist in the Drivers registry.

Payment Table

Reason: Any payment processed ensures the funds are directed securely toward an existing, valid rental agreement.

3. NOT NULL Constraint

Purpose

The NOT NULL constraint ensures that important fields always contain a value.

Applied Columns

Customers Table

Reason: Customer credentials and contact numbers can never be blank.

Vehicles Table

Reason: Without string representation, a rented car is meaningless on invoices.

Sales Table

Reason: Billing depends strictly on total calculations and timing durations.

4. UNIQUE Constraint

Purpose

The UNIQUE constraint ensures that duplicate values are not entered into a column.

Applied Columns

Drivers Table

Reason: Every scheduled driver must have a distinct driver's license ID.

Customers Table

Reason: Customers doing self-drives must provide driver's licenses preventing duplicates.

5. CHECK Constraint

Purpose

The CHECK constraint restricts values based on a specified condition.

Applied Columns

Expenses Table

Reason: Expenditure logging prevents users from mistakenly injecting negative amount.

Payments Table

Reason: Cash inflows generally cannot register with negative values.

Vehicles Table

Reason: To handle business share percentages properly for partial owning parameters.

6. DEFAULT Constraint

Purpose

The DEFAULT constraint automatically assigns a value when none is provided.

Applied Columns

Drivers Table

Reason: Every new driver onboarded inherently starts unassigned to a trip.

Expenses Table

Reason: Helps categorize standard fuel, maintenance unless 'Customer' is explicitly referenced.

7. AUTO_INCREMENT Constraint (MySQL alternative to IDENTITY)

Purpose

The AUTO_INCREMENT property automatically generates sequential numeric values for primary keys.

Applied Tables

- **Vehicles Table**
- **Sales Table**
- **Payment Table**
- **Expenses Table**

Reason: Let the system automatically issue distinct keys for transactions instead of manually determining IDs by the users.

8. Candidate Key Constraint

Purpose

A candidate key is a column or combination of columns that can uniquely identify a record.

Applied Examples

Customers Table

- customer_cnic
- license_number

Reason: Either unique identifier could identify the patron, though CNIC is used as the Primary Key.

9. Referential Integrity Constraint Summary

Purpose

Referential integrity checks verify relationships within the Rent-a-Car framework.

Applied Relationships in the System

- **Customers -> Sales**: Renting logs cannot outlive the customer making them.
- **Vehicles -> Sales**: No ghost cars. A rental must be correlated to the active fleet.
- **Sales -> Payments**: Incomes strictly associate with valid invoice bills (Sale IDs); stopping arbitrary fund associations.
- **Sales/Vehicles -> Expenses**: Explicit linking available if a certain ride directly accrued a refuel or a flat tire repair bill.

Benefits

- Eliminates database anomalies
- Reduces reporting biases on the Dashboard
- Guarantees robust financial auditing.

Conclusion

The Rent-a-Car Management System uses various SQL Server constraints including Primary Key, Foreign Key, Not Null, Unique, Check, Default, Identity, Composite Key, Candidate Key, and Referential Integrity constraints. These constraints ensure that only valid and consistent data is stored in the database, improving reliability, accuracy, and overall performance of the system. Proper implementation of these constraints is essential for maintaining a robust and efficient database environment.